

Digital Electronics Lab Report

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Experiment 7: BCD addition

Objectives:

(i) Add two BCD numbers to produce a valid BCD result.

Apparatus :

- IC-7483 – 4-bit adder (2)
- IC-7408 – Quad AND gate (1)
- IC-7432 – Quad OR gate (1)
- IC-7447 – BCD to 7-segment display decoder (1)
- 7-segment LED display – Common anode (1)
- Resistors – 470 Ω or 390 Ω (8)
- Resistors – 1 k Ω (4)

- Breadboard
- DIP Switch – With 4 or more switches (2)
- Connecting wires
- DC Power Supply – 5 V (1)

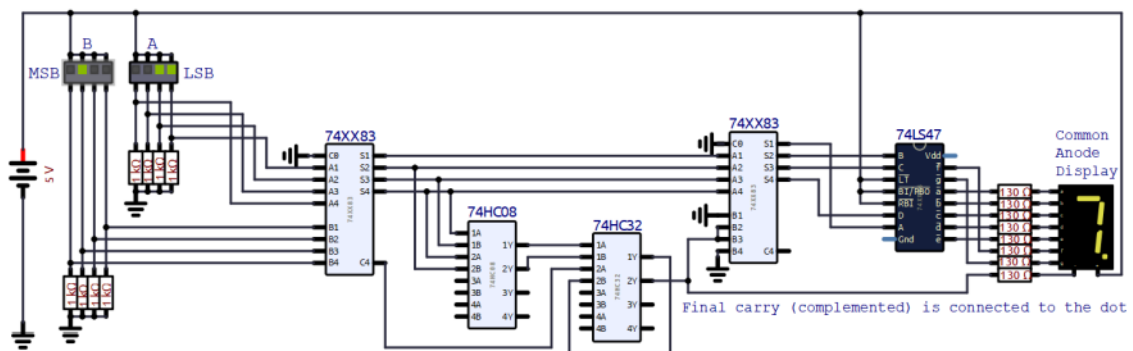
Theory :

Binary Coded Decimal (BCD) is a method of representing decimal digits where each digit from 0 to 9 is represented by its 4-bit binary equivalent (0000–1001). When two BCD digits are added using a standard binary adder such as the IC-7483, the resulting sum may sometimes be greater than 1001 (decimal 9), which is not a valid BCD representation.

Therefore, a correction must be applied to obtain a valid BCD result. The correction rule states that if the binary sum exceeds 1001 or if there is a carry generated from the 4-bit addition, the value 0110 (decimal 6) is added to the sum. This correction converts the invalid binary result into a valid BCD digit. In the BCD addition circuit, the first 7483 performs the binary addition of two BCD digits, while logic gates such as IC-7408 (AND) and IC-7432 (OR) detect whether the correction condition is satisfied. If required, a second 7483 adds 0110 to the intermediate sum to produce the correct BCD output. The final BCD result is then sent to a BCD-to-7-segment decoder such as IC-7447, which drives a 7-segment display to show the corresponding decimal digit. This method

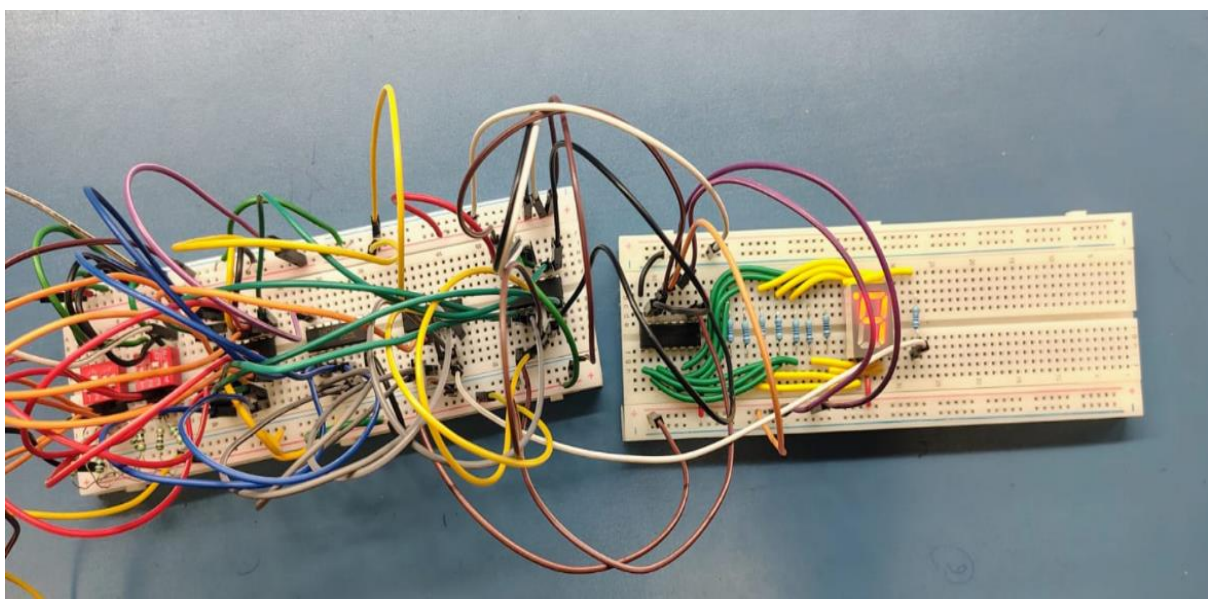
allows digital systems to correctly perform arithmetic operations on decimal numbers represented in BCD form.

Circuit Diagram :

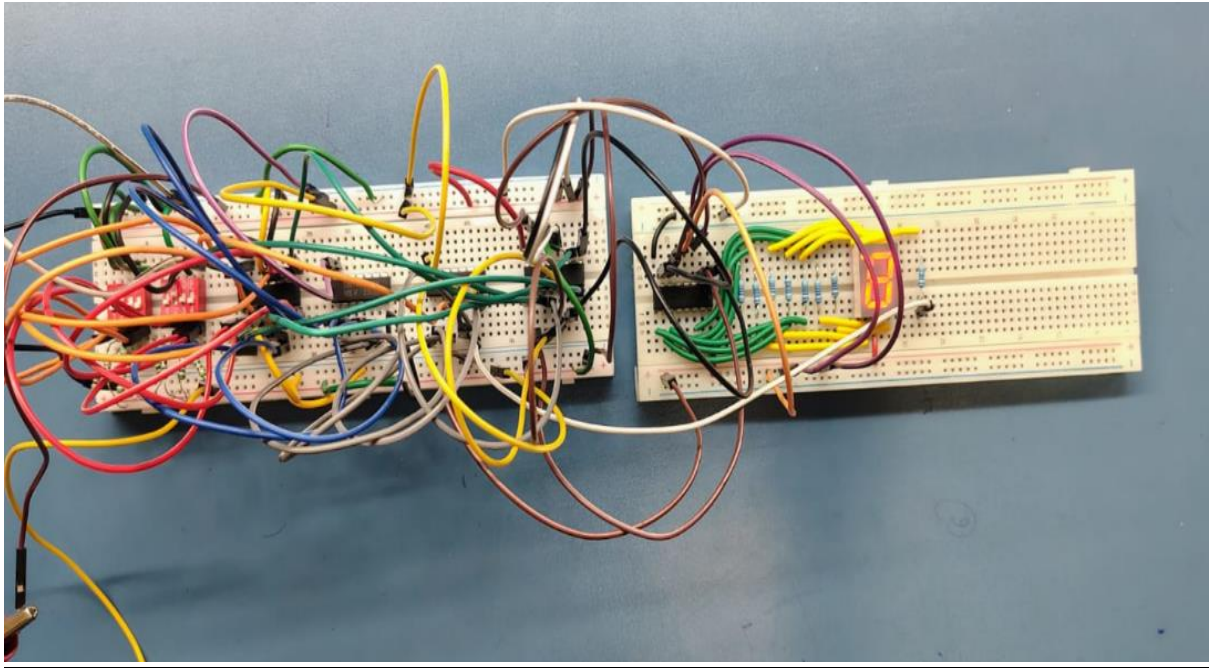


Observation :

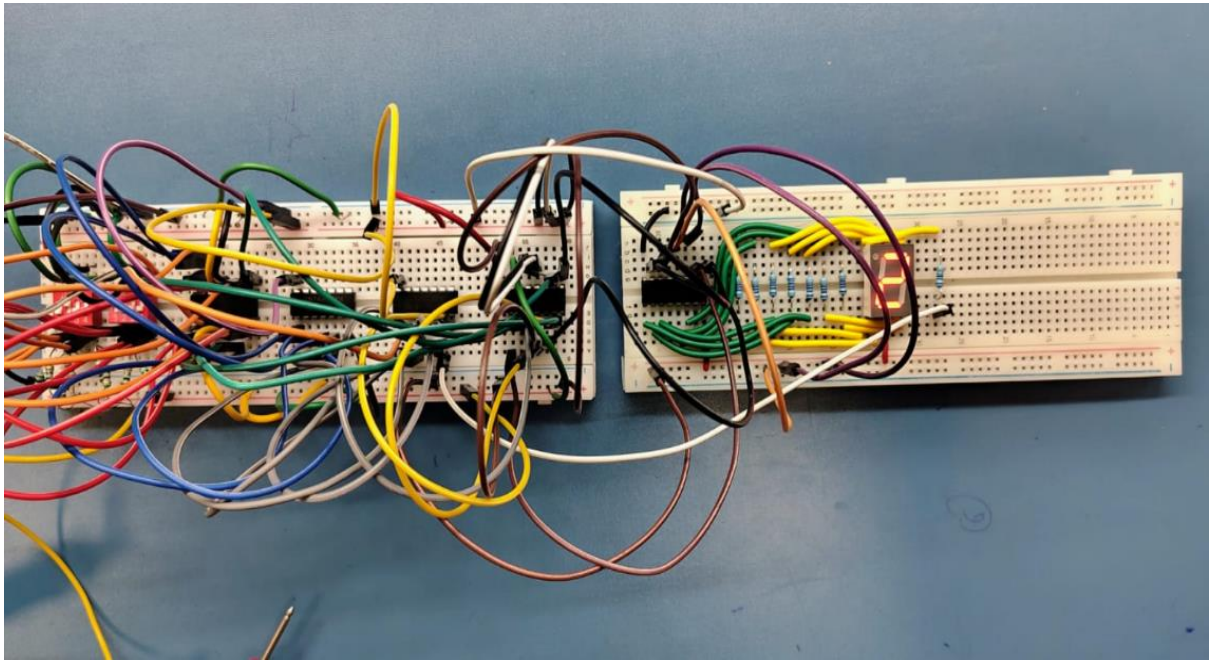
4+2:



4+4:



8+4:



Conclusion :

In this experiment, BCD addition was successfully implemented using digital logic ICs. Two BCD digits were first added using the 4-bit binary adder IC-7483, and a correction circuit using AND and OR gates was used to detect invalid BCD results. Whenever the binary sum exceeded 9 or a carry was generated, the correction value 0110 was added to obtain a valid BCD output. The final BCD result was then decoded using IC-7447 and displayed on a 7-segment display. Thus, the experiment verified the working principle of BCD addition and demonstrated how correction logic ensures proper decimal representation in digital systems.